Experiment 2 — Relational Schema & My SQL

Aim :

To convert the designed ER diagram into a relational schema and implemented it using MySQL with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE, and ON DELETE SET NULL constraints.

① Relational Schema

CUSTOMER

CUSTOMER(

CUSTOMER_ID PK,
First_Name,
Middle_Name,
Last_Name,
Email UNIQUE
DOB

)

CUSTOMER_PHONE

CUSTOMER_PHONE

Customer_ID PK, FK,
Phone PK

)

ADDRESS

Address_ID PK,
Customer_ID FK,
House_No,
Street,
Area,

City,
State,
PIN-Code,
Address_Type
)

SELLER

SELLER(
Seller_ID PK
Seller_Name
Email UNIQUE
Phone UNIQUE
)

CATEGORY

CATEGORY(
Category_ID PK,
Category_Name UNIQUE
Description
)

PRODUCT

PRODUCT(
Product_ID PK,
Product_Name,
Price,
Stock,
Description,
Seller_ID FK
Category_ID FK

ELECTRONICS

ELECTRONICS(

Product_ID PK, FK,

Warranty_Period

)

CLOTHING

CLOTHING(

Product_ID PK, FK,

Size,

Fabric

)

GROCERY

GROCERY(

Product_ID PK, FK,

~~Size~~ Expiry_Date

Weight

)

ORDERS

~~Order~~ ORDERS(

Order_ID PK,

Customer_ID FK,

Address_ID FK,

Order_Date,

Status,

Total_Amount

)

(2) My SQL Source Code

```
CREATE DATABASE Indion_Ecommerce;
USE Indion_Ecommerce;
CREATE TABLE Customer(
    Customer_ID INT PRIMARY KEY,
    First_Name VARCHAR(50) NOT NULL,
    Middle_Name VARCHAR(50),
    Last_Name VARCHAR(50) NOT NULL,
    Email VARCHAR(500) NOT NULL UNIQUE,
    DOB DATE
);
CREATE TABLE Customer Phone(
    Customer_ID INT PRIMARY KEY,
    Seller_Name VARCHAR(100) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Phone VARCHAR(15) UNIQUE
);
CREATE TABLE Seller(
    Seller_ID INT PRIMARY KEY,
    Seller_Name VARCHAR(100) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Phone VARCHAR(15) UNIQUE
);
```

```
CREATE TABLE Category (
Category_ID INT PRIMARY KEY,
Category_Name VARCHAR(100) NOT NULL UNIQUE,
Description VARCHAR(255)
);

CREATE TABLE Address (
    Address_ID INT PRIMARY KEY,
    Customer_ID INT NOT NULL,
    House_No VARCHAR(20) NOT NULL,
    Street VARCHAR(100) NOT NULL,
    Area VARCHAR(100),
    City VARCHAR(50) NOT NULL,
    State VARCHAR(50) NOT NULL,
    PIN_Code VARCHAR(10) NOT NULL,
    Address_Type VARCHAR(20) NOT NULL,

    FOREIGN KEY (Customer_ID)
        REFERENCES Customer (Customer_ID)
            ON DELETE CASCADE
    );
    CREATE TABLE Product (
        Product_ID INT PRIMARY KEY,
        Product_Name VARCHAR(100) NOT NULL,
        Price DECIMAL(10, 2) NOT NULL
```

```
    Stock INT NOT NULL,
    Description VARCHAR(255),
    Seller_ID INT NOT NULL,
    Category_ID INT NOT NULL,
    );
```

(3) Sample Data

Customer

```
INSERT INTO Customer
VALUES
(1, 'Dev', 'NULL', 'Parashar', dev@gmail.com,
                                '2004-08-19'),
(2, 'Ayush', 'NULL', 'Sharma', ayush@gmail.com,
                                '2005-07-21');
```

Customer Phone.

```
INSERT INTO Customer Phone
VALUES
(1, '8318813310'),
(2, '8318873938');
```

Seller

```
INSERT INTO Seller
VALUES
(101, 'Delhi Electronics', 'seller1@gmail.com',
                                '7300000201'),
(102, 'Fashion Hub', 'seller2@gmail.com',
                                '7300000002');
```

Category

```
INSERT INTO Category
VALUES
(201, 'Electronics', 'Electronic Product');
(202, 'Clothing', 'Fashion Product');
```

Address

```
INSERT INTO Address
VALUES
(301, 1, '12A', 'MG Road', 'Central Delhi',
 'Delhi', 'Delhi', '11000', 'Home');
```

(4) Demonstration ON DELETE CASCADE

```
DELETE FROM Orders
WHERE Order_ID = 501;
```

verify

```
SELECT * From Order Item;
SELECT * From Payment;
SELECT * FROM Delivery;
```

(5) Important Correction.

```
Category_ID INT
FOREIGN KEY (Category_ID)
    REFERENCES Category (Category_ID)
    ON DELETE SET NULL
```

(6) Output.

Query OK, 0 rows affected.